

1. Length of a Month

```
#include "library.h"
```

```
int length_of_a_month(const int month, const int year)
```

```
{
    if (month == 1) return 31;
    if (month == 2)
    {
        if (year % 4 == 0) return 29;
        else return 28;
    }
    if (month == 3) return 31;
    if (month == 4) return 30;
    if (month == 5) return 31;
    if (month == 6) return 30;
    if (month == 7) return 31;
    if (month == 8) return 31;
    if (month == 9) return 30;
    if (month == 10) return 31;
    if (month == 11) return 30;
    if (month == 12) return 31;
    int days = length_of_a_month(month,year);

}
```

```
int main()
```

```
{
    length_of_a_month(2,2008);
    int month = 2;
    int year = 2008;
    int days = length_of_a_month(month,year);
    {
        cout << "February " << year << " has " << days << " days." << endl;
    }
}
```

```
rabbiteng.miamledu - PuTTY
if (month == 2)
{
    if (year % 4 == 0) return 29;
    else return 28;
}
if (month == 3) return 31;
if (month == 4) return 30;
if (month == 5) return 31;
if (month == 6) return 30;
if (month == 7) return 31;
if (month == 8) return 31;
if (month == 9) return 30;
if (month == 10) return 31;
if (month == 11) return 30;
if (month == 12) return 31;
int days = length_of_a_month(month, year);
}

int main()
{
    length_of_a_month(2, 2008);
    int month = 2;
    int year = 2008;
    int days = length_of_a_month(month, year);
    {
        cout << "February " << year << " has " << days << " days." << endl;
    }
}

taai82@rabbit:~/ECE118/Labs/Lab6 % CC Lab6.cpp
Lab6.cpp: In function 'int length_of_a_month(int, int)':
Lab6.cpp:24:1: warning: control reaches end of non-void function [-Wreturn-type]
   24 | }
      | ^
taai82@rabbit:~/ECE118/Labs/Lab6 % a.out
February 2008 has 29 days.
taai82@rabbit:~/ECE118/Labs/Lab6 %
```

7 86°F
Mostly sunny

Search

Lab 6 Assignm rabbit.eng.miam Lab 6 - Microso rabbit.eng.mian

11:34 AM
10/1/2025

2. Day of the Year

```
#include "library.h"
#include <string>

int length_of_a_month(const int month, const int year)
{
    if (month == 1) return 31;
    if (month == 2)
    {
        if (year % 4 == 0) return 29;
        else return 28;
    }
    if (month == 3) return 31;
    if (month == 4) return 30;
    if (month == 5) return 31;
    if (month == 6) return 30;
    if (month == 7) return 31;
    if (month == 8) return 31;
    if (month == 9) return 30;
    if (month == 10) return 31;
    if (month == 11) return 30;
    if (month == 12) return 31;
    int days = length_of_a_month(month, year);

}

int day_of_the_year(const int day, const int month, const int year)
{
    int total_days = 0;
    for (int m = 1; m < month; m++)
    {
        total_days += length_of_a_month(m, year);
    }
    total_days += day;
    return total_days;
}

string month_name(int month)
{
    if (month == 1) return "January";
    if (month == 2) return "February";
```

```

        if (month == 3) return "March";
        if (month == 4) return "April";
        if (month == 5) return "May";
        if (month == 6) return "June";
        if (month == 7) return "July";
        if (month == 8) return "August";
        if (month == 9) return "September";
        if (month == 10) return "October";
        if (month == 11) return "November";
        if (month == 12) return "December";
    }

```

```

string day_suffix(int day)
{
    if (day % 10 == 1 && day != 11) return "st";
    if (day % 10 == 2 && day != 12) return "nd";
    if (day % 10 == 3 && day != 13) return "rd";
    return "th";
}

```

```

int main()
{
    int day = 30;
    int month = 9;
    int year = 2025;
    int total = day_of_the_year(30,9,2025);

    cout << "The " << day << day_suffix(day) << " of " << month_name(month) << " is the "
    << total << day_suffix(total) << " day of the year." << endl;
}

```

```
rabbit.eng.miamiedu - PuTTY
if (month == 1) return "January";
if (month == 2) return "February";
if (month == 3) return "March";
if (month == 4) return "April";
if (month == 5) return "May";
if (month == 6) return "June";
if (month == 7) return "July";
if (month == 8) return "August";
if (month == 9) return "September";
if (month == 10) return "October";
if (month == 11) return "November";
if (month == 12) return "December";
}

string day_suffix(int day)
{
    if (day % 10 == 1 && day != 11) return "st";
    if (day % 10 == 2 && day != 12) return "nd";
    if (day % 10 == 3 && day != 13) return "rd";
    return "th";
}

int main()
{
    int day = 30;
    int month = 9;
    int year = 2025;
    int total = day_of_the_year(30,9,2025);

    cout << "The " << day << day_suffix(day) << " of " << month_name(month) << " is the " << total << day_suffix(total) << " day of the year." << endl;
}

taai82@rabbit:~$ cd Lab6.cpp
Lab6.cpp: In function 'int length_of_a_month(int, int)':
Lab6.cpp:24:1: warning: control reaches end of non-void function [-Wreturn-type]
  24 | }
      | ^
Lab6.cpp: In function 'std::string month_name(int)':
Lab6.cpp:53:1: warning: control reaches end of non-void function [-Wreturn-type]
   53 | }
      | ^
taai82@rabbit:~$ a.out
The 30th of September is the 273rd day of the year.
taai82@rabbit:~$
```

78°F
Mostly sunny

Search

rabbit.eng.miamiedu Lab 6 Assignment rabbit.eng.miamiedu Lab 6 - Microsoft

12:20 PM
10/1/2025

3. Day of the Century

```
#include "library.h"
#include <string>

int length_of_a_month(const int month, const int year)
{
    if (month == 1) return 31;
    if (month == 2)
    {
        if (year % 4 == 0) return 29;
        else return 28;
    }
    if (month == 3) return 31;
    if (month == 4) return 30;
    if (month == 5) return 31;
    if (month == 6) return 30;
    if (month == 7) return 31;
    if (month == 8) return 31;
    if (month == 9) return 30;
    if (month == 10) return 31;
    if (month == 11) return 30;
    if (month == 12) return 31;
    int days = length_of_a_month(month, year);

}

int day_of_the_year(const int day, const int month, const int year)
{
    int total_days = 0;
    for (int m = 1; m < month; m++)
    {
        total_days += length_of_a_month(m, year);
    }
    total_days += day;
    return total_days;
}

string month_name(int month)
{
    if (month == 1) return "January";
    if (month == 2) return "February";
```

```

        if (month == 3) return "March";
        if (month == 4) return "April";
        if (month == 5) return "May";
        if (month == 6) return "June";
        if (month == 7) return "July";
        if (month == 8) return "August";
        if (month == 9) return "September";
        if (month == 10) return "October";
        if (month == 11) return "November";
        if (month == 12) return "December";
    }

```

```

string day_suffix(int day)
{
    if (day % 10 == 1 && day != 11) return "st";
    if (day % 10 == 2 && day != 12) return "nd";
    if (day % 10 == 3 && day != 13) return "rd";
    return "th";
}

```

```

int day_of_the_century(const int day, const int month, const int year, const int century)
{
    int total_days_of_century = 0;
    for (int d = 1; d < month; d++)
    {
        total_days_of_century += day_of_the_year(d, month, year);
    }
    total_days_of_century += day;
    return total_days_of_century;
}

```

```

int main()
{
    int day = 30;
    int month = 9;
    int year = 2025;
    int century = 2000-2099;
    int total = day_of_the_century(30,9,2025,century);

    cout << "The " << day << day_suffix(day) << " of " << month_name(month) << " " <<
year << " is the " << total << day_suffix(total) << " day of the century." << endl;
}

```

```
rabbiteng.mianieda - PuTTY
string day_suffix(int day)
{
    if (day % 10 == 1 && day != 11) return "st";
    if (day % 10 == 2 && day != 12) return "nd";
    if (day % 10 == 3 && day != 13) return "rd";
    return "th";
}

int day_of_the_century(const int day, const int month, const int year, const int century)
{
    int total_days_of_century = 0;
    for (int d = 1; d < month; d++)
    {
        total_days_of_century += day_of_the_year(d, month, year);
    }
    total_days_of_century += day;
    return total_days_of_century;
}

int main()
{
    int day = 30;
    int month = 9;
    int year = 2025;
    int century = 2000-2099;
    int total = day_of_the_century(30, 9, 2025, century);

    cout << "The " << day << day_suffix(day) << " of " << month_name(month) << " " << year << " is the " << total << day_suffix(total) << " day of the century." << endl;
}

taa182@rabbit:~/ECE118/Labs/Lab6 $ CC Lab6.cpp
Lab6.cpp: In function 'int length_of_a_month(int, int)':
Lab6.cpp:24:1: warning: control reaches end of non-void function [-Wreturn-type]
   24 | }
      | ^
Lab6.cpp: In function 'std::string month_name(int)':
Lab6.cpp:53:1: warning: control reaches end of non-void function [-Wreturn-type]
   53 | }
      | ^
taa182@rabbit:~/ECE118/Labs/Lab6 $ a.out
The 30th of September 2025 is the 2010th day of the century.
taa182@rabbit:~/ECE118/Labs/Lab6 $
```


4. Day of Forever

```
#include "library.h"
```

```
#include <string>
```

```
int length_of_a_month(const int month, const int year)
```

```
{
    if (month == 1) return 31;
    if (month == 2)
    {
        if (year % 4 == 0) return 29;
        else return 28;
    }
    if (month == 3) return 31;
    if (month == 4) return 30;
    if (month == 5) return 31;
    if (month == 6) return 30;
    if (month == 7) return 31;
    if (month == 8) return 31;
    if (month == 9) return 30;
    if (month == 10) return 31;
    if (month == 11) return 30;
    if (month == 12) return 31;
    const int days = length_of_a_month(month, year);

}
```

```
int day_of_the_year(const int day, const int month, const int year)
```

```
{
    int total_days = 0;
    for (int m = 1; m < month; m++)
    {
        total_days += length_of_a_month(m, year);
    }
    total_days += day;
    return total_days;
}
```

```
string month_name(int month)
```

```
{
    if (month == 1) return "January";
```

```

        if (month == 2) return "February";
        if (month == 3) return "March";
        if (month == 4) return "April";
        if (month == 5) return "May";
        if (month == 6) return "June";
        if (month == 7) return "July";
        if (month == 8) return "August";
        if (month == 9) return "September";
        if (month == 10) return "October";
        if (month == 11) return "November";
        if (month == 12) return "December";
    }

```

```

string day_suffix(int day)
{
    if (day % 10 == 1 && day != 11) return "st";
    if (day % 10 == 2 && day != 12) return "nd";
    if (day % 10 == 3 && day != 13) return "rd";
    return "th";
}

```

```

int day_of_the_century(const int day, const int month, const int year, const int century)
{
    int total_days_of_century = 0;
    for (int d = 1; d < month; d++)
    {
        total_days_of_century += day_of_the_year(d, month, year);
    }
    total_days_of_century += day;
    return total_days_of_century;
}

```

```

int day_of_forever(const int day, const int month, const int year)
{
    int total_days = 0;

    for (int y = 0; y < year; y++)
    {
        if (y % 400 == 0) total_days += 366;
        else if (y % 100 == 0) total_days += 365;
        else if (y % 4 == 0) total_days += 366;
    }
}

```

```

        else total_days += 365;
    }
    total_days += day_of_the_year(day, month, year);
    return total_days;
}

int main()
{
    int day = 30;
    int month = 9;
    int year = 2025;

    int total = day_of_forever(day, month, year);

    cout << "The " << day << day_suffix(day) << " of " << month_name(month) << " " <<
year << " was day number " << total << "." << endl;

    //sample testing//
    cout << "The 1st of January 2000 is day number " << day_of_forever(1,1,2000) << endl;
    cout << "The 1st of January 1900 is day number " << day_of_forever(1,1,1900) << endl;
    cout << "The 4th of July 1776 is day number " << day_of_forever(4,7,1776) << endl;
    cout << "The 1st of October 2024 is day number " << day_of_forever(1,10,2024) <<
endl;
    cout << "The 7th of October 2024 is day number " << day_of_forever(7,10,2024) <<
endl;
    cout << "The 27th of November 2737 is day number " << day_of_forever(27,11,2737) <<
endl;
    cout << "The 1st of January 10 A.D. is day number " << day_of_forever(1,1,10) << endl;
}

```

```
rabbiteng@miamiedu: ~ - PuTTY
for (int y = 0; y < year; y++)
{
    if (y % 400 == 0) total_days += 366;
    else if (y % 100 == 0) total_days += 365;
    else if (y % 4 == 0) total_days += 366;
    else total_days += 365;
}

total_days += day_of_the_year(day, month, year);
return total_days;
}

int main()
{
    int day = 30;
    int month = 9;
    int year = 2025;

    int total = day_of_forever(day, month, year);

    cout << "The " << day << day_suffix(day) << " of " << month_name(month) << " " << year << " was day number " << total << "." << endl;

    //sample testing//
    cout << "The 1st of January 2000 is day number " << day_of_forever(1,1,2000) << endl;
    cout << "The 1st of January 1900 is day number " << day_of_forever(1,1,1900) << endl;
    cout << "The 4th of July 1776 is day number " << day_of_forever(4,7,1776) << endl;

    taa182@rabbit:~/ECE118/Labs/Lab6 & CC Lab6.cpp
    Lab6.cpp: In function 'int length_of_a_month(int, int)':
    Lab6.cpp:24:1: warning: control reaches end of non-void function [-Wreturn-type]
       24 | }
          | ^
    Lab6.cpp: In function 'std::string month_name(int)':
    Lab6.cpp:53:1: warning: control reaches end of non-void function [-Wreturn-type]
       53 | }
          | ^
    taa182@rabbit:~/ECE118/Labs/Lab6 & a.out
    The 30th of September 2025 was day number 739890.
    The 1st of January 2000 is day number 730486
    The 1st of January 1900 is day number 693962
    The 4th of July 1776 is day number 648857
    The 1st of October 2024 is day number 739526
    The 7th of October 2024 is day number 739532
    The 27th of November 2737 is day number 1000000
    The 1st of January 10 A.D. is day number 3654
    taa182@rabbit:~/ECE118/Labs/Lab6 & [ ]
```

5. Day of the Week

```
#include "library.h"
```

```
#include <string>
```

```
int length_of_a_month(const int month, const int year)
```

```
{
    if (month == 1) return 31;
    if (month == 2)
    {
        if (year % 4 == 0) return 29;
        else return 28;
    }
    if (month == 3) return 31;
    if (month == 4) return 30;
    if (month == 5) return 31;
    if (month == 6) return 30;
    if (month == 7) return 31;
    if (month == 8) return 31;
    if (month == 9) return 30;
    if (month == 10) return 31;
    if (month == 11) return 30;
    if (month == 12) return 31;
    const int days = length_of_a_month(month, year);

}
```

```
int day_of_the_year(const int day, const int month, const int year)
```

```
{
    int total_days = 0;
    for (int m = 1; m < month; m++)
    {
        total_days += length_of_a_month(m, year);
    }
    total_days += day;
    return total_days;
}
```

```
string month_name(int month)
```

```
{
    if (month == 1) return "January";
```

```

        if (month == 2) return "February";
        if (month == 3) return "March";
        if (month == 4) return "April";
        if (month == 5) return "May";
        if (month == 6) return "June";
        if (month == 7) return "July";
        if (month == 8) return "August";
        if (month == 9) return "September";
        if (month == 10) return "October";
        if (month == 11) return "November";
        if (month == 12) return "December";
    }

```

```

string day_suffix(int day)
{
    if (day % 10 == 1 && day != 11) return "st";
    if (day % 10 == 2 && day != 12) return "nd";
    if (day % 10 == 3 && day != 13) return "rd";
    return "th";
}

```

```

int day_of_the_century(const int day, const int month, const int year, const int century)
{
    int total_days_of_century = 0;
    for (int d = 1; d < month; d++)
    {
        total_days_of_century += day_of_the_year(d, month, year);
    }
    total_days_of_century += day;
    return total_days_of_century;
}

```

```

int day_of_forever(const int day, const int month, const int year)
{
    int total_days = 0;

    for (int y = 1; y < year; y++)
    {
        if (y % 400 == 0) total_days += 366;
        else if (y % 100 == 0) total_days += 365;
        else if (y % 4 == 0) total_days += 366;
    }
}

```

```

        else total_days += 365;
    }

    total_days += day_of_the_year(day, month, year);
    return total_days;
}

int day_of_the_week(const int day, const int month, const int year)
{
    const int week = day_of_forever(day, month, year);
    return week % 7;
}

string week_name(int week)
{
    if (week == 0) return "Sunday";
    if (week == 1) return "Monday";
    if (week == 2) return "Tuesday";
    if (week == 3) return "Wednesday";
    if (week == 4) return "Thursday";
    if (week == 5) return "Friday";
    if (week == 6) return "Saturday";
    return "";
}

int main()
{
    int day = 30;
    int month = 9;
    int year = 2025;
    int week = day_of_the_week(day, month, year);

    cout << "The " << day << day_suffix(day) << " of " << month_name(month) << " " <<
    year << " was a " << week_name(week) << "." << endl;

    //sample testing//
    cout << "The 1st of January 2000 was a " << week_name(day_of_the_week(1,1,2000))
    << "." << endl;
    cout << "The 1st of January 1900 was a " << week_name(day_of_the_week(1,1,1900))
    << "." << endl;
}

```

```

        cout << "The 4th of July 1776 was a " << week_name(day_of_the_week(4,7,1776)) <<
"." << endl;
        cout << "The 1st of October 2024 was a " << week_name(day_of_the_week(1,10,2024))
<< "." << endl;
        cout << "The 7th of October 2024 was a " << week_name(day_of_the_week(7,10,2024))
<< "." << endl;
        cout << "The 27th of November 2737 will be a " <<
week_name(day_of_the_week(27,11,2737)) << "." << endl;
        cout << "The 1st of January 10 A.D. was a " << week_name(day_of_the_week(1,1,10))
<< "." << endl;
}

```

The screenshot shows a Windows terminal window titled "rabbit.eng.miamiedu - PuTTY". The terminal displays the following C++ code and its output:

```

cout << "The 1st of October 2024 was a " << week_name(day_of_the_week(1,10,2024)) << "." << endl;
cout << "The 7th of October 2024 was a " << week_name(day_of_the_week(7,10,2024)) << "." << endl;
cout << "The 27th of November 2737 will be a " << week_name(day_of_the_week(27,11,2737)) << "." << endl;
cout << "The 1st of January 10 A.D. was a " << week_name(day_of_the_week(1,1,10)) << "." << endl;
}

taai82@rabbit:~/ECE110/Labs/Lab6 $ CC Lab6.cpp
Lab6.cpp: In function 'int length_of_a_month(int, int)':
Lab6.cpp:25:1: warning: control reaches end of non-void function [-Wreturn-type]
   25 | }
      | ^
Lab6.cpp: In function 'std::string month_name(int)':
Lab6.cpp:54:1: warning: control reaches end of non-void function [-Wreturn-type]
   54 | }
      | ^
taai82@rabbit:~/ECE110/Labs/Lab6 $ a.out
The 30th of September 2025 was a Tuesday.
The 1st of January 2000 was a Saturday.
The 1st of January 1900 was a Monday.
The 4th of July 1776 was a Thursday.
The 1st of October 2024 was a Tuesday.
The 7th of October 2024 was a Monday.
The 27th of November 2737 will be a Saturday.
The 1st of January 10 A.D. was a Friday.
taai82@rabbit:~/ECE110/Labs/Lab6 $

```

The terminal window also shows the Windows taskbar at the bottom with various application icons and a system clock indicating 7:12 PM on 10/1/2025.

6. A Calendar for a Month

```
#include "library.h"
```

```
#include <string>
```

```
int length_of_a_month(const int month, const int year)
```

```
{
    if (month == 1) return 31;
    if (month == 2)
    {
        if (year % 4 == 0) return 29;
        else return 28;
    }
    if (month == 3) return 31;
    if (month == 4) return 30;
    if (month == 5) return 31;
    if (month == 6) return 30;
    if (month == 7) return 31;
    if (month == 8) return 31;
    if (month == 9) return 30;
    if (month == 10) return 31;
    if (month == 11) return 30;
    if (month == 12) return 31;
    const int days = length_of_a_month(month, year);

}
```

```
int day_of_the_year(const int day, const int month, const int year)
```

```
{
    int total_days = 0;
    for (int m = 1; m < month; m++)
    {
        total_days += length_of_a_month(m, year);
    }
    total_days += day;
    return total_days;
}
```

```
string month_name(int month)
```

```
{
    if (month == 1) return "January";
```

```

        if (month == 2) return "February";
        if (month == 3) return "March";
        if (month == 4) return "April";
        if (month == 5) return "May";
        if (month == 6) return "June";
        if (month == 7) return "July";
        if (month == 8) return "August";
        if (month == 9) return "September";
        if (month == 10) return "October";
        if (month == 11) return "November";
        if (month == 12) return "December";
    }

```

```

string day_suffix(int day)
{
    if (day % 10 == 1 && day != 11) return "st";
    if (day % 10 == 2 && day != 12) return "nd";
    if (day % 10 == 3 && day != 13) return "rd";
    return "th";
}

```

```

int day_of_the_century(const int day, const int month, const int year, const int century)
{
    int total_days_of_century = 0;
    for (int d = 1; d < month; d++)
    {
        total_days_of_century += day_of_the_year(d, month, year);
    }
    total_days_of_century += day;
    return total_days_of_century;
}

```

```

int day_of_forever(const int day, const int month, const int year)
{
    int total_days = 0;

    for (int y = 1; y < year; y++)
    {
        if (y % 400 == 0) total_days += 366;
        else if (y % 100 == 0) total_days += 365;
        else if (y % 4 == 0) total_days += 366;
    }
}

```

```

        else total_days += 365;
    }

    total_days += day_of_the_year(day, month, year);
    return total_days;
}

int day_of_the_week(const int day, const int month, const int year)
{
    const int week = (day_of_forever(day, month, year) + 6) % 7;
    return week;
}

string week_name(int week)
{
    if (week == 0) return "Mon";
    if (week == 1) return "Tue";
    if (week == 2) return "Wed";
    if (week == 3) return "Thu";
    if (week == 4) return "Fri";
    if (week == 5) return "Sat";
    if (week == 6) return "Sun";
    return "";
}

void draw_calendar(int month, int year)
{
    cout << " -----" << endl;
    cout << "|          " << month_name(month) << " " << year << "      |" << endl;
    cout << " -----" << endl;
    cout << "|Mon Tue Wed Thu Fri Sat Sun|" << endl;

    int first_day_of_the_week = day_of_the_week(1, month, year);
    int days = length_of_a_month(month, year);
    int number_of_days = 0;

    cout << " -----" << endl;

    for (int i = 0; i < first_day_of_the_week; i++)
    {

```

```
    cout << "|  ";  
    number_of_days++;  
}
```

```
for (int d = 1; d <= days; d++)  
{  
    if (d < 10) cout << "| " << d << " ";  
    else cout << "| " << d << " ";  
    number_of_days++;  
  
    if (number_of_days % 7 == 0)  
    {  
        cout << "|" << endl;  
        cout << "-----" << endl;  
    }  
}
```

```
if (number_of_days % 7 != 0)  
{  
    int remaining = 7 - (number_of_days % 7);  
    for (int i = 0; i < remaining; i++)  
        cout << "|  ";  
    cout << "|" << endl;  
    cout << "-----" << endl;  
}  
}
```

```
int main()  
{  
    draw_calendar(10,2025);  
}
```

```
rabbiteng.miamiedu - PuTTY

taa182@rabbit:~/ECE118/Labs/REDONE_Lab6$ a.out
a.out: Command not found.
taa182@rabbit:~/ECE118/Labs/REDONE_Lab6$ CC REDONE_Lab6.cpp
REDONE_Lab6.cpp: In function 'int length_of_a_month(int, int)':
REDONE_Lab6.cpp:25:1: warning: control reaches end of non-void function [-Wreturn-type]
   25 | }
      | ^
REDONE_Lab6.cpp: In function 'std::string month_name(int)':
REDONE_Lab6.cpp:54:1: warning: control reaches end of non-void function [-Wreturn-type]
   54 | }
      | ^
taa182@rabbit:~/ECE118/Labs/REDONE_Lab6$ a.out
-----
|          October 2025          |
| Mon Tue Wed Thu Fri Sat Sun |
|-----|
|   |   |   | 1 | 2 | 3 | 4 | 5 |
|-----|
| 6 | 7 | 8 | 9 |10 |11 |12 |
|-----|
|13 |14 |15 |16 |17 |18 |19 |
|-----|
|20 |21 |22 |23 |24 |25 |26 |
|-----|
|27 |28 |29 |30 |31 |   |   |
|-----|
taa182@rabbit:~/ECE118/Labs/REDONE_Lab6$
```

7. A Solid Product

```
#include "library.h"
```

```
#include <string>
```

```
int length_of_a_month(const int month, const int year)
```

```
{
    if (month == 1) return 31;
    if (month == 2)
    {
        if (year % 4 == 0) return 29;
        else return 28;
    }
    if (month == 3) return 31;
    if (month == 4) return 30;
    if (month == 5) return 31;
    if (month == 6) return 30;
    if (month == 7) return 31;
    if (month == 8) return 31;
    if (month == 9) return 30;
    if (month == 10) return 31;
    if (month == 11) return 30;
    if (month == 12) return 31;
    const int days = length_of_a_month(month, year);

}
```

```
int day_of_the_year(const int day, const int month, const int year)
```

```
{
    int total_days = 0;
    for (int m = 1; m < month; m++)
    {
        total_days += length_of_a_month(m, year);
    }
    total_days += day;
    return total_days;
}
```

```
string month_name(int month)
```

```
{
    if (month == 1) return "January";
```

```

    if (month == 2) return "February";
    if (month == 3) return "March";
    if (month == 4) return "April";
    if (month == 5) return "May";
    if (month == 6) return "June";
    if (month == 7) return "July";
    if (month == 8) return "August";
    if (month == 9) return "September";
    if (month == 10) return "October";
    if (month == 11) return "November";
    if (month == 12) return "December";
}

```

```

string day_suffix(int day)
{
    if (day % 10 == 1 && day != 11) return "st";
    if (day % 10 == 2 && day != 12) return "nd";
    if (day % 10 == 3 && day != 13) return "rd";
    return "th";
}

```

```

int day_of_the_century(const int day, const int month, const int year, const int century)
{
    int total_days_of_century = 0;
    for (int d = 1; d < month; d++)
    {
        total_days_of_century += day_of_the_year(d, month, year);
    }
    total_days_of_century += day;
    return total_days_of_century;
}

```

```

int day_of_forever(const int day, const int month, const int year)
{
    int total_days = 0;

    for (int y = 1; y < year; y++)
    {
        if (y % 400 == 0) total_days += 366;
        else if (y % 100 == 0) total_days += 365;
        else if (y % 4 == 0) total_days += 366;
    }
}

```

```

        else total_days += 365;
    }

    total_days += day_of_the_year(day, month, year);
    return total_days;
}

int day_of_the_week(const int day, const int month, const int year)
{
    const int week = (day_of_forever(day, month, year) + 6) % 7;
    return week;
}

string week_name(int week)
{
    if (week == 0) return "Mon";
    if (week == 1) return "Tue";
    if (week == 2) return "Wed";
    if (week == 3) return "Thu";
    if (week == 4) return "Fri";
    if (week == 5) return "Sat";
    if (week == 6) return "Sun";
    return "";
}

void draw_calendar(int month, int year)
{
    cout << " -----" << endl;
    cout << "|          " << month_name(month) << " " << year << "      |" << endl;
    cout << " -----" << endl;
    cout << "|Mon Tue Wed Thu Fri Sat Sun|" << endl;

    int first_day_of_the_week = day_of_the_week(1, month, year);
    int days = length_of_a_month(month, year);
    int number_of_days = 0;

    cout << " -----" << endl;

    for (int i = 0; i < first_day_of_the_week; i++)
    {

```



```
    cout << "|  ";
    number_of_days++;
}
```

```
for (int d = 1; d <= days; d++)
{
    if (d < 10) cout << "| " << d << " ";
    else cout << "| " << d << " ";
    number_of_days++;

    if (number_of_days % 7 == 0)
    {
        cout << "|" << endl;
        cout << "-----" << endl;
    }
}
```

```
if (number_of_days % 7 != 0)
{
    int remaining = 7 - (number_of_days % 7);
    for (int i = 0; i < remaining; i++)
        cout << "|  ";
    cout << "|" << endl;
    cout << "-----" << endl;
}
}
```

```
int main()
{
    draw_calendar(2,2304);
}
```

```
rabbiteng@miamiledu - PuTTY

    cout << "-----" << endl;
}

if (number_of_days % 7 != 0)
{
    int remaining = 7 - (number_of_days % 7);
    for (int i = 0; i < remaining; i++)
        cout << "|   ";
    cout << "|" << endl;
    cout << "-----" << endl;
}

int main()
{
    draw_calendar(2,2304);
}

taa182@rabbit:~/ECE118/Labs/REDONE_Lab6 % CC REDONE_Lab6.cpp
REDONE_Lab6.cpp: In function 'int length_of_a_month(int, int)':
REDONE_Lab6.cpp:25:1: warning: control reaches end of non-void function [-Wreturn-type]
  25 | }
     | ^
REDONE_Lab6.cpp: In function 'std::string month_name(int)':
REDONE_Lab6.cpp:54:1: warning: control reaches end of non-void function [-Wreturn-type]
  54 | }
     | ^
taa182@rabbit:~/ECE118/Labs/REDONE_Lab6 % a.out
-----
|           February 2304           |
| Mon Tue Wed Thu Fri Sat Sun |
|  1  2  3  4  5  6  7  |
|  8  9 10 11 12 13 14  |
| 15 16 17 18 19 20 21  |
| 22 23 24 25 26 27 28  |
| 29 |   |   |   |   |   |
taa182@rabbit:~/ECE118/Labs/REDONE_Lab6 %
```

And for November 2246...

```
rabbiteng@miamiledu - PuTTY

    cout << "-----" << endl;
}

if (number_of_days % 7 != 0)
{
    int remaining = 7 - (number_of_days % 7);
    for (int i = 0; i < remaining; i++)
        cout << "|   ";
    cout << "|" << endl;
    cout << "-----" << endl;
}

int main()
{
    draw_calendar(11,2246);
}

taa182@rabbit:~/ECE118/Labs/REDONE_Lab6 % CC REDONE_Lab6.cpp
REDONE_Lab6.cpp: In function 'int length_of_a_month(int, int)':
REDONE_Lab6.cpp:25:1: warning: control reaches end of non-void function [-Wreturn-type]
  25 | }
     | ^
REDONE_Lab6.cpp: In function 'std::string month_name(int)':
REDONE_Lab6.cpp:54:1: warning: control reaches end of non-void function [-Wreturn-type]
  54 | }
     | ^
taa182@rabbit:~/ECE118/Labs/REDONE_Lab6 % a.out
-----
|           November 2246           |
| Mon Tue Wed Thu Fri Sat Sun |
|   |   |   |   |   | 1 |
|  2  3  4  5  6  7  8  |
|  9 10 11 12 13 14 15  |
| 16 17 18 19 20 21 22  |
| 23 24 25 26 27 28 29  |
| 30 |   |   |   |   |   |
taa182@rabbit:~/ECE118/Labs/REDONE_Lab6 %
```